Practice: one consumer, five budgets

Lecture notes

What this lecture is about. Lecture 11 was a practice lecture. It worked through one question in five parts. The consumer starts with the utility function $u = xy$ and a simple budget line. In each part one thing changes: both prices, one price, a price that rises after the first units (a kinked budget line), and finally the preferences themselves. Each part is solved with the same method, set out below.

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1 The method

Marginal utility per euro ("bang for the buck")

MU_x is the marginal utility of x: the derivative of $u(x, y)$ with respect to x, with y held fixed. The ratio MU_x/p_x is the extra utility from the last euro spent on x. In class this ratio was called the *bang for the buck* on x.

At any bundle on a budget line, compare the two ratios:

$$\frac{\mathrm{MU}_x}{p_x} \quad \text{versus} \quad \frac{\mathrm{MU}_y}{p_y}.$$

- If the ratio is higher for x, moving one euro from y to x raises utility: move along the line towards more x.
- If the ratio is higher for y, move towards less x.
- If the two ratios are equal, no small move helps: stop.

With a kinked budget line, each segment has its own prices, so the comparison uses the prices of the segment one is on. The steps are always the same: (i) on each segment, find the point where the two ratios are equal; (ii) check whether that point lies on the segment; (iii) if it does not, follow the direction of the comparison to the end of the segment; (iv) at a kink, compare the ratios with the prices just before and just after it.

The pictures. In every picture, x is on the horizontal axis and y on the vertical axis. The budget line is orange and the budget set is shaded; a thin dashed orange line is the budget line of the previous part, or the continuation of a segment. The blue curve is the indifference curve through the best bundle. The arrows show the direction in which the comparison pushes. A label such as "at $(0, 8)$: 8 vs 0" means that at the bundle $(0, 8)$ the ratio is 8 for x and 0 for y. With two segments, the label starts with the segment number.

2 Part 1: one budget line

Question, part 1

A consumer has the utility function $u = xy$. Both prices are €1 and the income is €8, so the budget line is $x + y = 8$. Find the best affordable bundle.

Step 1: marginal utilities. $\mathrm{MU}_x = y$ and $\mathrm{MU}_y = x$.

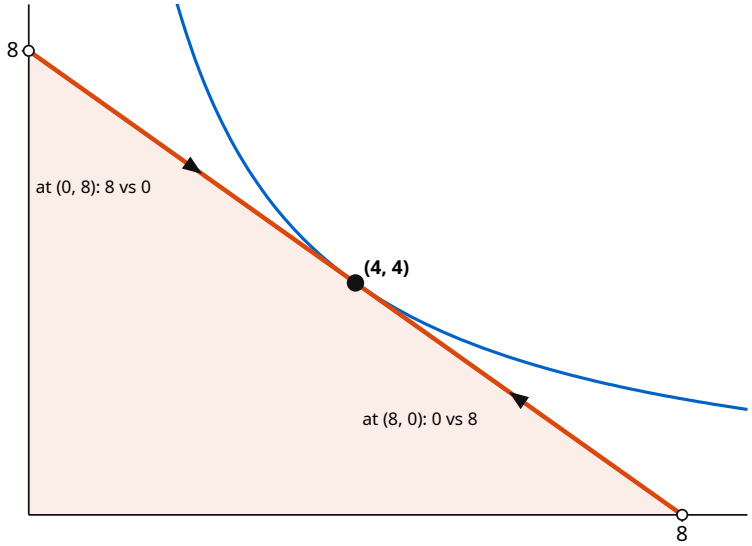
Step 2: equal ratios. With $p_x = p_y = 1$:

$$\frac{y}{1} = \frac{x}{1} \quad \Longrightarrow \quad y = x.$$

Step 3: the budget line. Substituting $y = x$ into $x + y = 8$ gives $2x = 8$, so $x = 4$ and $y = 4$. This point lies on the line.

Step 4: the direction elsewhere. At $(0, 8)$ the ratios are 8 for x against 0 for y, so the consumer moves towards more x. At $(8, 0)$ they are 0 against 8, so the consumer moves towards less x. Both directions lead to $(4, 4)$, the only point where the ratios are equal, so no comparison of utility levels is needed.

The best bundle is $(4, 4)$, with $u = 4 \cdot 4 = 16$.



Part 1: the budget line $x + y = 8$ and the best bundle $(4, 4)$.

3 Part 2: both prices double

Question, part 2

Both prices double, to €2 each; the income stays €8. The budget line is $2x + 2y = 8$, that is $x + y = 4$ after dividing by 2. Find the new best bundle.

Step 1: marginal utilities. Unchanged: $\mathrm{MU}_x = y$ and $\mathrm{MU}_y = x$.

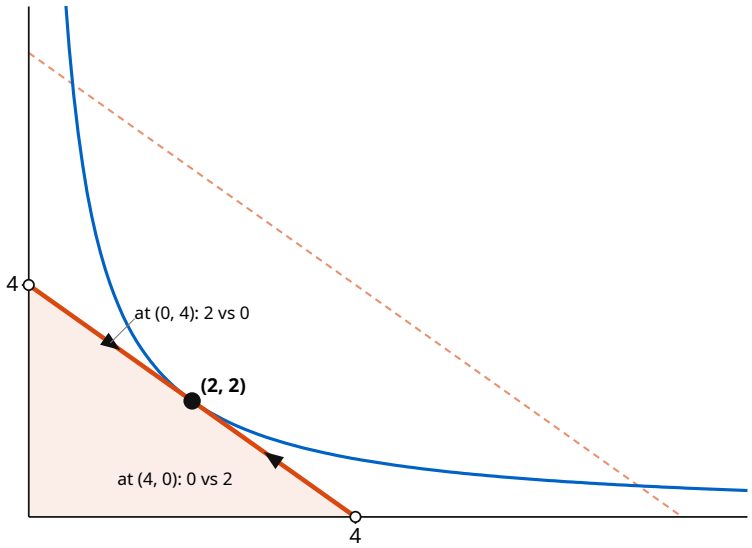
Step 2: equal ratios. With $p_x = p_y = 2$:

$$\frac{y}{2} = \frac{x}{2} \quad \Longrightarrow \quad y = x.$$

Step 3: the budget line. Substituting into $x + y = 4$ gives $2x = 4$, so $x = 2$ and $y = 2$.

Step 4: the direction elsewhere. At $(0, 4)$: $4/2 = 2$ for x against $0/2 = 0$ for y, so more x. At $(4, 0)$: 0 against $4/2 = 2$, so less x. The only stopping point is $(2, 2)$.

The best bundle is $(2, 2)$, with $u = 4$. Utility falls from 16 to 4: doubling both prices with the same income halves the budget line.



Part 2: the budget line $x + y = 4$ and the best bundle $(2, 2)$.

4 Part 3: the price of x falls

Question, part 3

Starting from part 2, the price of x falls from €2 to €1; the price of y stays €2 and the income €8. The budget line is $x + 2y = 8$. Find the new best bundle.

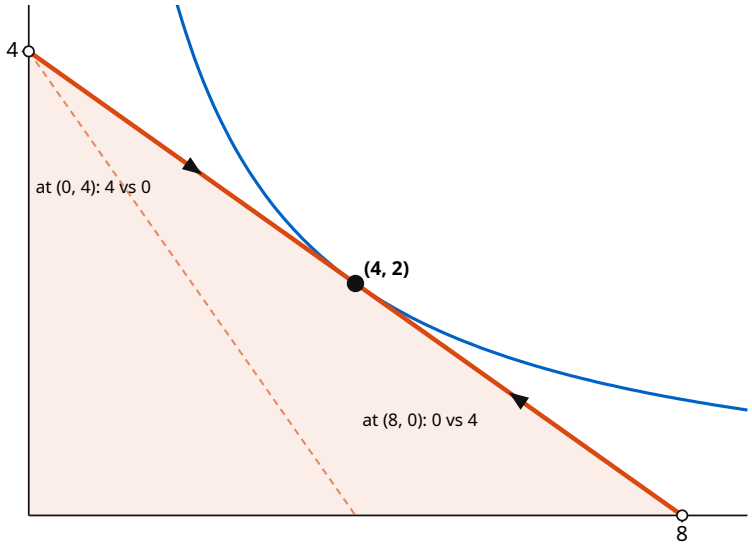
Step 1: equal ratios. With $p_x = 1$ and $p_y = 2$:

$$\frac{y}{1} = \frac{x}{2} \quad \Longrightarrow \quad y = \tfrac{1}{2}x.$$

Step 2: the budget line. Substituting into $x + 2y = 8$ gives $x + x = 2x = 8$, so $x = 4$ and $y = 2$.

Step 3: the direction elsewhere. At $(0, 4)$: $4/1 = 4$ for x against $0/2 = 0$ for y, so more x. At $(8, 0)$: 0 against $8/2 = 4$, so less x. The only stopping point is $(4, 2)$.

The best bundle is $(4, 2)$, with $u = 8$. Utility rises from 4 to 8. The consumer buys twice as much x as in part 2 and the same amount of y.



Part 3: the budget line $x + 2y = 8$ and the best bundle $(4, 2)$.

5 Part 4: a kinked budget line

Question, part 4

The prices are as in part 3 ($p_x = 1$, $p_y = 2$, income €8), but after the first 2 units of x, the price of x rises from €1 to €3. Find the budget line and the new best bundle.

Step 1: the budget line. Up to $x = 2$ the old line applies: $x + 2y = 8$. The first 2 units cost €2, which leaves €6; each further unit of x costs €3, so $3(x - 2) + 2y = 6$, that is $3x + 2y = 12$. The budget line has two segments:

$$x + 2y = 8 \ \text{ for } 0 \le x \le 2, \qquad 3x + 2y = 12 \ \text{ for } 2 \le x \le 4.$$

Both segments pass through the kink $(2, 3)$.

Step 2: segment 1, $x + 2y = 8$. Equal ratios, as in part 3, give $y = \frac{1}{2}x$ and the point $(4, 2)$. This point is *off* the segment, which runs only from $x = 0$ to $x = 2$. At $(0, 4)$ the ratios are 4 for x against 0 for y, so more x: along the whole segment the consumer moves right, to the kink $(2, 3)$.

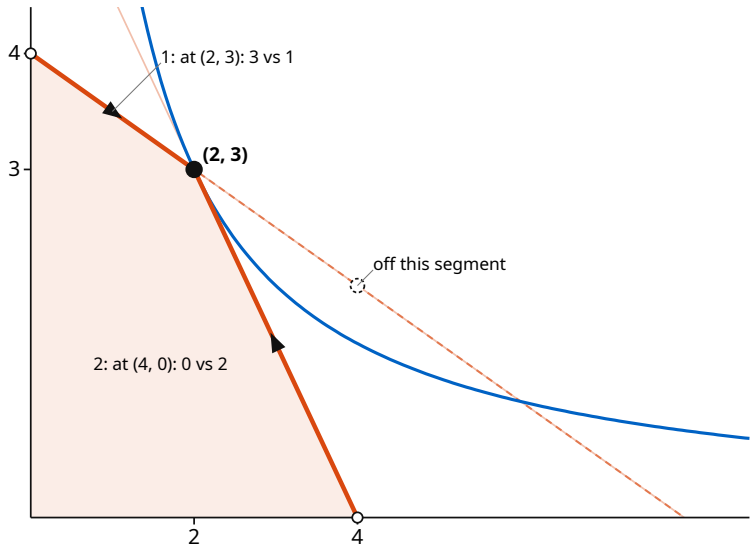
Step 3: segment 2, $3x + 2y = 12$. With $p_x = 3$ and $p_y = 2$:

$$\frac{y}{3} = \frac{x}{2} \quad \Longrightarrow \quad y = \tfrac{3}{2}x.$$

Substituting gives $3x + 3x = 6x = 12$, so $x = 2$ and $y = 3$: the kink itself, which lies on this segment. At $(4, 0)$ the ratios are $0/3 = 0$ for x against $4/2 = 2$ for y, so less x: the consumer moves left, to $(2, 3)$.

Step 4: the kink $(2, 3)$. Just before it, with prices $(1, 2)$: $3/1 = 3$ for x against $2/2 = 1$ for y, so more x. Just after it, with prices $(3, 2)$: $3/3 = 1$ against $2/2 = 1$, equal. The consumer stops at the kink.

The best bundle is the kink $(2, 3)$, with $u = 6$. Utility falls from 8 to 6.



Part 4: the kinked budget line and the best bundle $(2, 3)$. The dashed circle marks the point $(4, 2)$ from segment 1, which lies off that segment.

6 Part 5: new preferences

Question, part 5

The budget line is as in part 4. The consumer's preferences change to $u = xy^2$. Find the best bundle.

Step 1: marginal utilities. $\text{MU}_x = y^2$ and $\text{MU}_y = 2xy$.

Step 2: segment 1, $x + 2y = 8$**.** With prices $(1, 2)$:

$$\frac{y^2}{1} = \frac{2xy}{2} \quad \Longrightarrow \quad y = x.$$

Substituting gives $3x = 8$, so $x = y = \frac{8}{3}$. Since $\frac{8}{3} > 2$, this point is off the segment. At $(0, 4)$ the ratios are 16 for x against 0 for y, so more x: the consumer moves right, to $(2, 3)$.

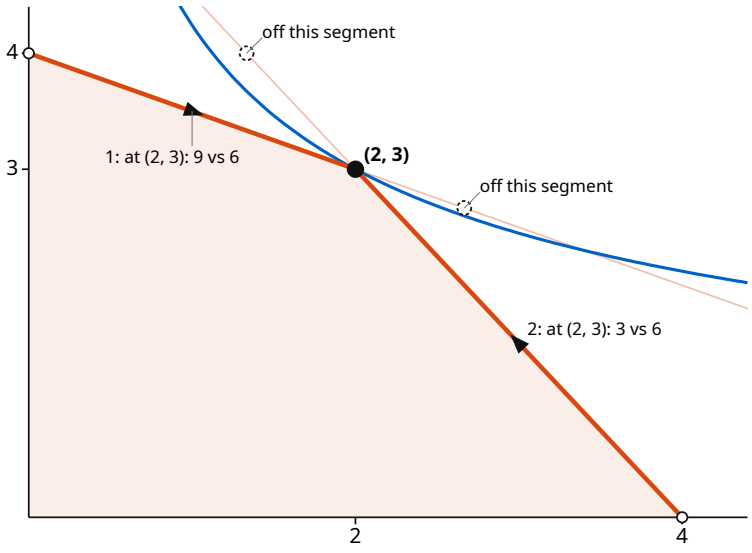
Step 3: segment 2, $3x + 2y = 12$**.** With prices $(3, 2)$:

$$\frac{y^2}{3} = \frac{2xy}{2} \quad \Longrightarrow \quad y = 3x.$$

Substituting gives $3x + 6x = 9x = 12$, so $x = \frac{4}{3}$ and $y = 4$. Since $\frac{4}{3} < 2$, this point is off the segment, which runs from $x = 2$ to $x = 4$. At $(2, 3)$ the ratios are $9/3 = 3$ for x against $12/2 = 6$ for y, so less x: along this segment the consumer moves left, to $(2, 3)$.

Step 4: the kink $(2, 3)$**.** Here $\text{MU}_x = 9$ and $\text{MU}_y = 12$. Just before the kink, with prices $(1, 2)$: 9 against 6, so more x. Just after it, with prices $(3, 2)$: 3 against 6, so less x. The consumer stops at the kink.

The best bundle is again the kink $(2, 3)$, now with $u = 2 \cdot 3^2 = 18$. The best bundle does not move. The utility function is new, so its value (18) cannot be compared with the value 6 of part 4.



Part 5: the same kinked budget line with $u = xy^2$. Both equal-ratio points lie off their segments; the best bundle is the kink $(2, 3)$.

Insight: why the kink is so often the answer

At a kink the price of x jumps. A comparison can point towards more x at the lower price and towards less x at the higher price. Then the consumer stops at the kink, even if the two ratios are not equal there. A range of different preferences can therefore all lead to the same kink, as parts 4 and 5 show.

7 Summary

part	utility	budget	best bundle	utility level
1	xy	$x + y = 8$	$(4, 4)$	16
2	xy	$x + y = 4$	$(2, 2)$	4
3	xy	$x + 2y = 8$	$(4, 2)$	8
4	xy	kinked at $(2, 3)$	$(2, 3)$	6
5	xy^2	kinked at $(2, 3)$	$(2, 3)$	18

- On a single budget line with $u = xy$, the best bundle is where $\text{MU}_x/p_x = \text{MU}_y/p_y$.
- Doubling both prices with the same income (part 2) moves the budget line inwards in parallel; with $u = xy$ the consumer keeps the same proportions but buys less of both.
- With a kinked budget line, solve each segment with its own prices, check whether the equal-ratio point lies on the segment, and compare the ratios on both sides of the kink.
- Utility levels can be compared only for the same utility function.